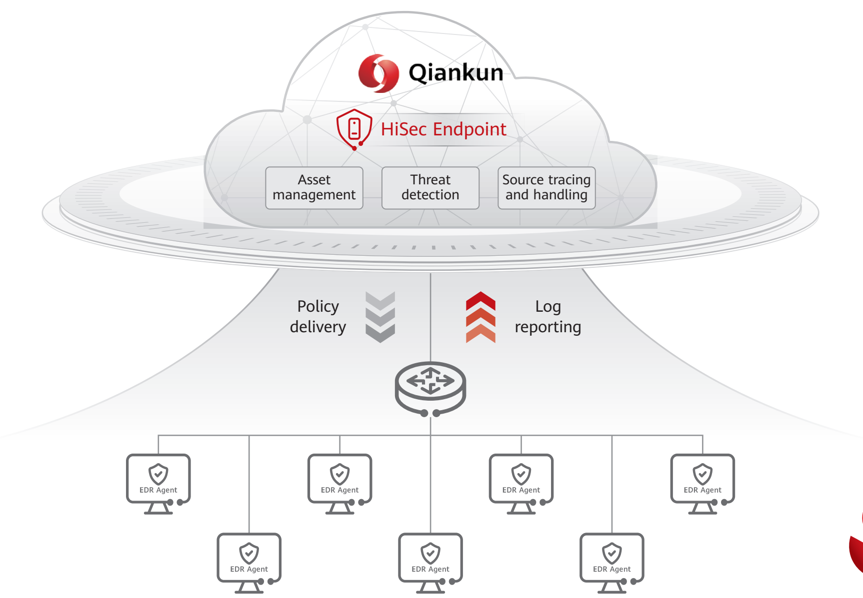
Huawei HiSec Endpoint

With client software installed on enterprise endpoints, Huawei HiSec Endpoint can detect abnormal endpoint behaviors in real time and identify potential threats from multiple dimensions. It can handle threat events promptly, isolate malicious files, terminate malicious processes, and scan for malware. By automatically tracing intrusion paths, HiSec Endpoint quickly evaluates the reach of attacks and performs in-depth malware scan for comprehensive risk hardening. Moreover, it supports Network Access Control for endpoint security, strictly manages endpoint access, ensuring that only compliant devices can access the enterprise network and protecting the security of enterprise core endpoint assets.

Highlights of HiSec Endpoint are as follows:

- Real-time security situational awareness
- One-click scan for malware and quick threat eradication
- Quick setup of endpoint protection capabilities
- NAC supports dynamic compliance security checks and real-time threat assessments to ensure compliant access of endpoints

Endpoint Detection and Response Solution



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As digitalization spreads to every industry, enterprises are increasingly dependent on network communications technologies to accommodate their business needs. In addition, enterprises, with multiple PCs and servers, are facing increasing network security risks. This is well solved by HiSec Endpoint , a service that detects and handles risks on local endpoints of enterprises to prevent virus infection and threat spreading on the intranet. By simply deploying the lightweight software HiSec Endpoint on their endpoints, enterprises can be fortified with security in seconds.

HiSec Endpoint comes with an antivirus engine and a behavior detection engine. Based on the full attack path detection rules, HiSec Endpoint can detect files and directories on endpoints just in milliseconds to quickly identify threats. For ransomware, HiSec Endpoint adopts the bait capture technology to discern risks at the early stage of malware intrusion and report abnormal events to the cloud in time. In addition, HiSec Endpoint synchronizes threat information in real time and updates the threat signature database immediately after detecting new threats. This can boost the network-wide security.

Highlights

Intelligent antivirus engine, detection rate of unknown threats >95%

- Huawei's self-developed third-generation AI-powered intelligent anti-virus engine. It supports rapid and accurate identification of file types and in-depth content analysis, and can recognize semantic content and hidden malicious behaviors. The MDL detection engine enables programmable virus detection, covering hundreds of millions of viruses. The neural network AI engine, based on Huawei's SiteAI platform and neural network algorithms, supports the detection of unknown threats. The detection rate for unknown variant viruses is >95%.

100% detection rate of known ransomware

- Offers leading advanced ransomware detection, static analysis of key memory scan, and dynamic simulation based on Emulator, implementing 100% detection rate of known ransomware, and 95% detection precision of unknown ransomware variants. Provides unique technology of recovering ransomware-encrypted file, innovative tactics of backing up patented kernel, and automatic learning of ransomware encryption behavior, achieving real-time blocking of ransomware malicious processes, file backup in seconds, and zero data loss.

95% detection precision rate of zero-day exploits

- Detects fileless attacks and vulnerability exploits based on user-mode and kernel-mode Shellcode detection, in-depth memory analysis, process behavior baseline, advanced injection detection, and instruction sequence, and implements precise analysis based on the memory-based graph engine to detect unknown fileless threats with the precision rate of zero-day exploits over 95%.

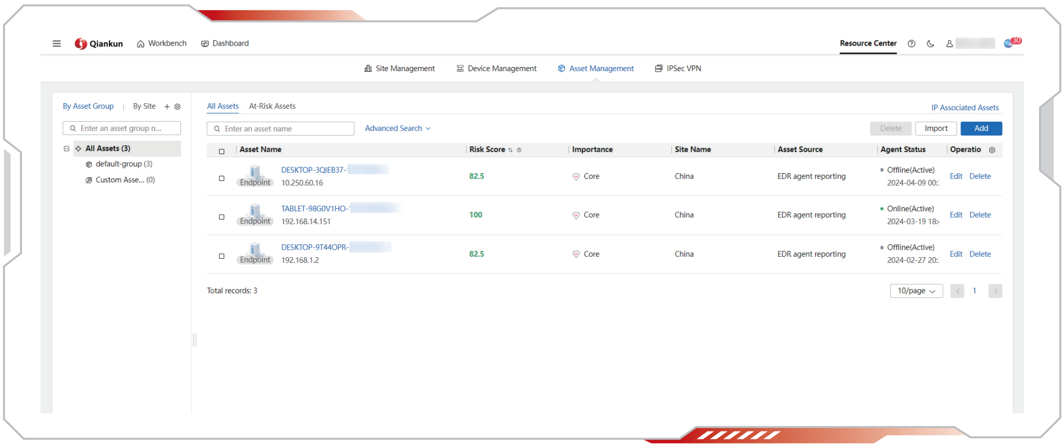
90% blocking rate of phishing Trojan horses

- Phishing Trojan horses are usually sent to victims preset by attack organizations through emails. Compared with common malicious programs, phishing Trojan horses are easier to lure victims to execute. HiSec Endpoint precisely captures the inducing features of phishing Trojan horses based on big data and promptly blocks Trojan horses before they affect the system, achieving the blocking rate over 90%.

Key Feature

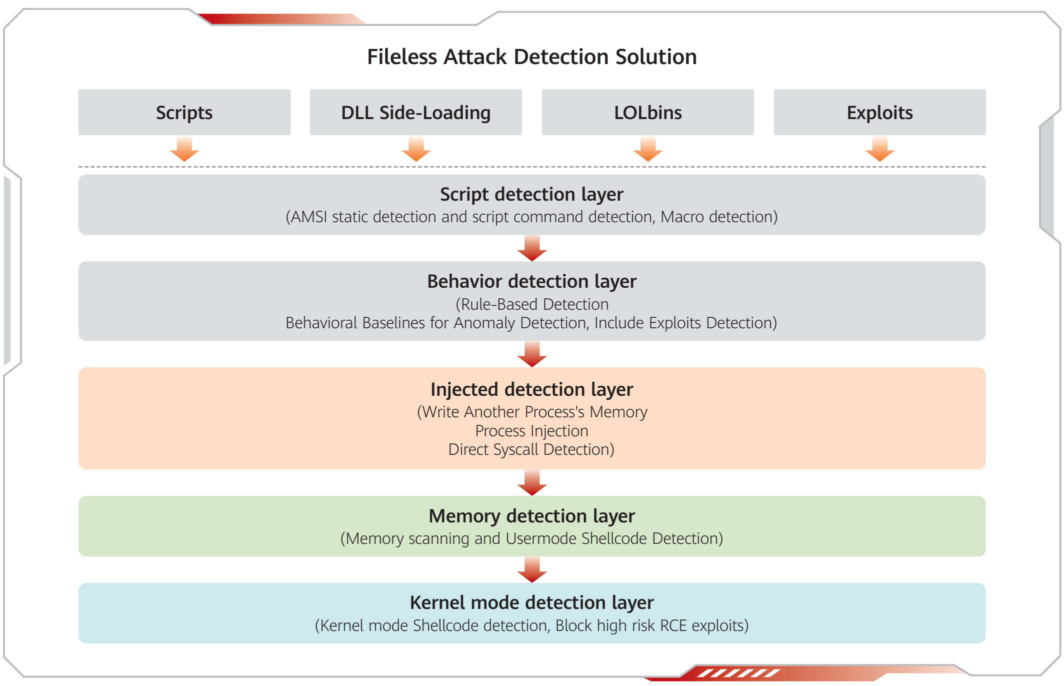
(1) Asset visibility: real-time asset status detection

HiSec Endpoint ensures automatic inventory on endpoint assets so that assets can be automatically identified and uniformly managed. HiSec Endpoint intelligently analyzes the security status of endpoint assets and displays the analysis scores and risk overview.



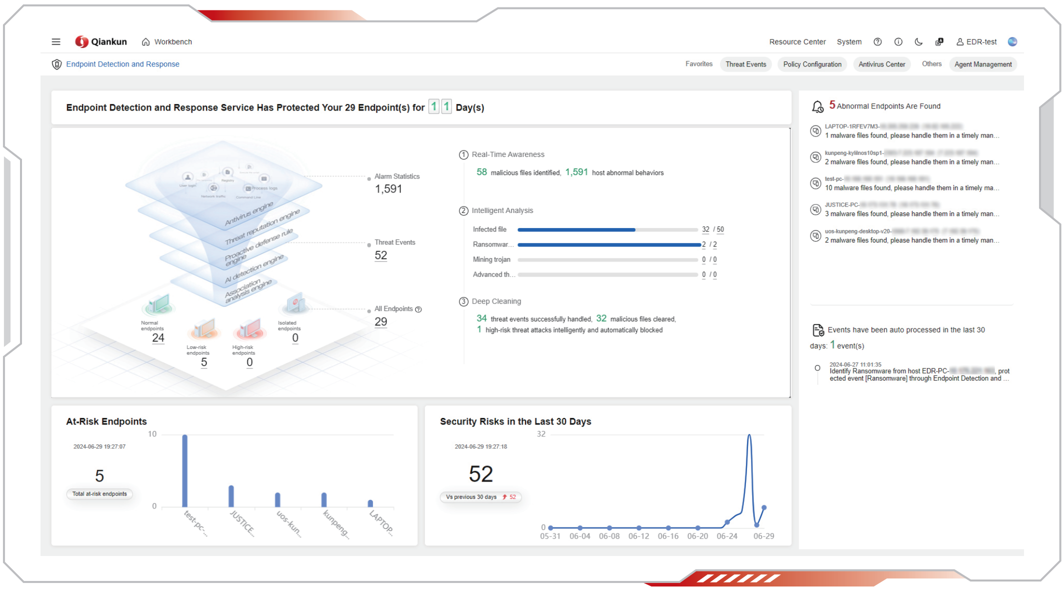
(2) A multi-layer protection system to block various advanced threats

HiSec Endpoint uses lightweight endpoint Agent to block system behaviors such as APIs, processes, networks, and files of endpoints. Based on the unique memory-based graph engine, HiSec Endpoint analyzes the behaviors of the entire attack path to precisely detect and associate fileless attacks and vulnerabilities. Finally, the entire attack path is displayed in the memory-based graph engine, facilitating event response.



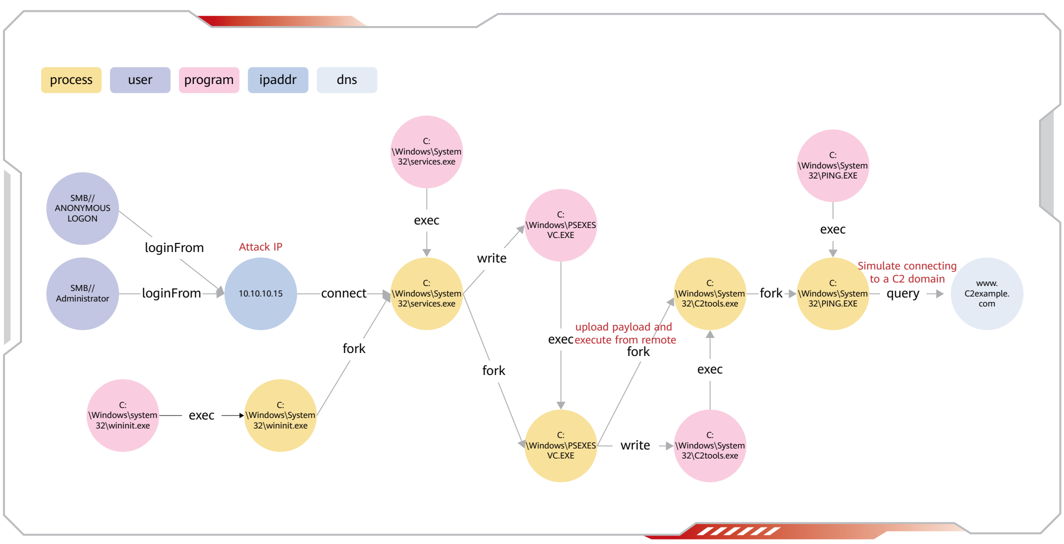
(3) Threat visualization: overall security situation display

HiSec Endpoint's management platform is built on Huawei Qiankun Unified Threat Analysis and Management Platform. This platform supports multiple security services like perimeter protection, threat intelligence, and network threat assessment. Relying on Huawei Qiankun's unified security operation center, it natively enables centralized storage, retrieval, unified analysis, and visualization of security logs. It has a unified cross-product management background and policy configuration, and supports XDR cross-domain threat judgment and one-click automated response via cross-domain correlation algorithms.



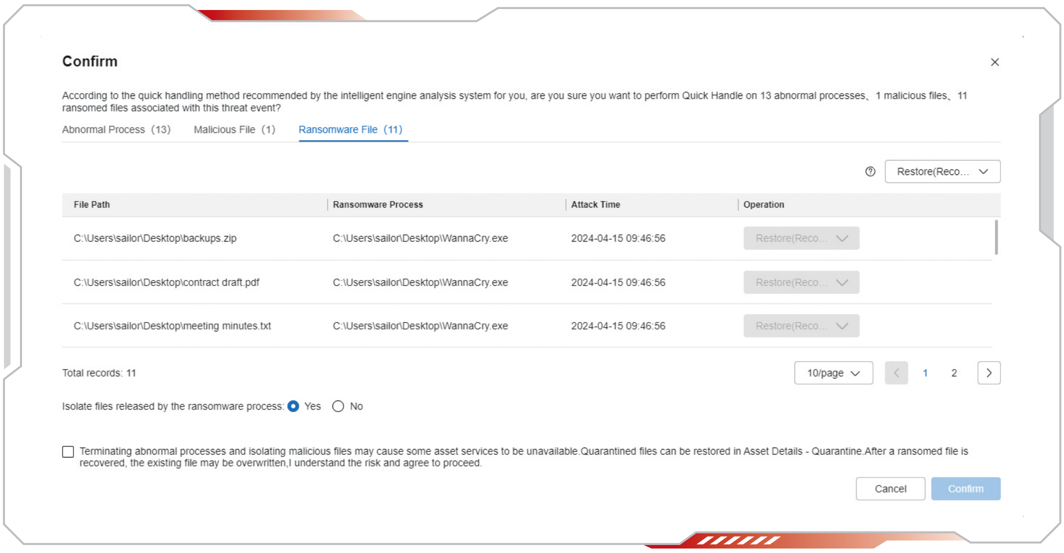
(4) Attack visualization: proactive source tracing and forensics for IOC

By leveraging the offline storage of source tracing data building on endpoint-cloud collaboration and the high-performance provenance graph database on the cloud, HiSec Endpoint supports one-click global source tracing for the specified IOC and automatically displays attack paths of more than 10 hops within 1s. Supported IOC types include domain name, IP, program hash, file name, process name, registry path, etc.



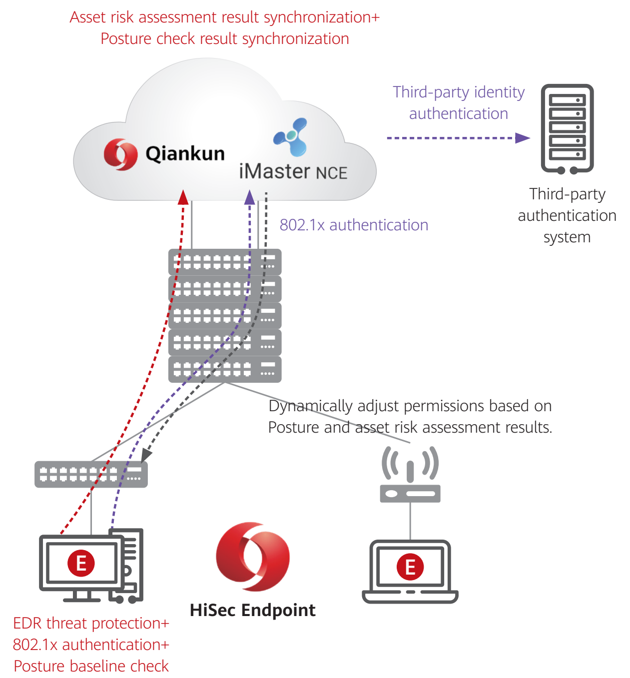
(5) Ransomware rollback, one-click recovery of files encrypted by ransomware, zero data loss

Event-triggered backup, detecting file exceptions at the kernel level in real time and backs up files. One-click handling and rollback based on ransomware events, ensuring data security.



Network Access Control Solution

In campus security access scenarios,
802.1x+Posture baseline check+threat prevention is supported



Key Feature

(1) Dynamic Compliance Security Checks

Supports 17 Posture baseline checks, including those related to operating systems, patches, anti-virus status, processes, and passwords, to ensure the security of endpoints accessing the network.

(2) Dynamic Real-time Threat Assessments

Supports dynamic threat assessment associated with network access control. It enables network device isolation, security device blocking of access, micro-isolation by host firewalls, and 6 types of automatic repairs for non-compliance issues.

(3) Endpoint-Network Linkage for Threat Containment across the Network

Based on two dimensions-the results of Posture baseline checks and the comprehensive risk scores of threat protection, it makes comprehensive judgments for authorization. Once a threat is detected, it can interwork with switches and APs to isolate endpoints in seconds.

(4) Unified Access via One Platform

With a unified controller and a unified agent, and unified policies and identities, it provides a consistent experience and reduces Opex by 70%.

Specifications

(1) Endpoint Detection and Response

Category	Level-1 Specification	Level-2 Specification	Standard edition			Professional edition		
			Windows	Linux	macOS	Windows	Linux	macOS
Endpoint Management	Asset Management	Endpoint Collection and Identification	√	√	√	√	√	√
		Historical IP Association	√	√	√	√	√	X
		Asset Registration	√	X	X	√	√	X
		Scan Code Registration	√	X	X	√	√	X
		Asset Discovery	X	X	X	X	√	X
	Endpoint Agent Installation and Deployment	Local Download and Installation	√	√	√	√	√	√
		Email Promotion Installation	√	√	√	√	√	√
		Batch Installation via Domain Control	√	√	√	√	√	√
		Silent Installation	√	√	X	√	√	X
	Endpoint Agent Upgrade	Automatic Upgrade	√	√	√	√	√	√
		Manual Upgrade	√	√	√	√	√	√
	Endpoint Agent Uninstallation	Endpoint Uninstallation Code Uninstallation	√	√	√	√	√	√
		Administrator Cloud Uninstallation	√	√	√	√	√	√
	Virus Database Distribution	Offline Virus Database Manual Import and Distribution	√	√	√	√	√	√
		Online Virus Database/HIPS/Agent Synchronization	√	√	√	√	√	√
		Active Virus Database Upgrade Push	√	√	√	√	√	√

Category	Level-1 Specification	Level-2 Specification	Standard edition			Professional edition		
			Windows	Linux	macOS	Windows	Linux	macOS
		Incremental Upgrade	√	√	√	√	√	√
		Virus Database Gradual Upgrade	√	√	√	√	√	√
	Policy Management	Batch Policy Issuance	√	√	√	√	√	√
		Message Policy Configuration	√	√	√	√	√	√
		Multi-tenancy	√	√	√	√	√	√
		Account Permission Hierarchical Management	√	√	√	√	√	√
	Endpoint Security Visibility	Endpoint Threat Event Visibility	√	√	√	√	√	√
		Endpoint Quarantine Area Management	√	√	√	√	√	√
		Endpoint Process Management	X	X	X	√	√	√
Active Defense	Black and White Lists	Process Blacklist	√	√	√	√	√	√
		File Whitelist	√	√	√	√	√	√
		Self-learning Whitelist	X	X	X	√	√	√
	Bait Capture	Bait Capture	√	√	X	√	√	√
	Backup and Recovery	Ransomware Local Lightweight Backup and Recovery	√	X	X	√	X	√
	File Anti-tampering	File Anti-tampering	X	X	X	√	X	X
	Network Access Policy	Host Firewall	√	√	X	√	√	X
		Single Network Connectivity	√	√	X	√	√	√
Detection	Virus Detection	Online Scan and Clean	√	√	√	√	√	√
		Scan and Clean Mode	√	√	√	√	√	√
		Scan Policy	√	√	√	√	√	√
		Real-time Protection	√	√	√	√	√	√
		Periodic Scan and Clean	√	√	√	√	√	√
		USB Drive Scan and Clean	√	√	X	√	√	√
		Cloud-based Scan and Clean	X	X	X	X	X	X
		Virus Types	√	√	√	√	√	√
	Mining Detection	Mining Trojan Intelligence-based Detection	X	X	X	√	√ (Kernel-mode) X (User-mode)	X
		Mining Trojan Behavior Detection	√	√	X	√	√	X
	Data Theft Detection	Data Theft Trojan Detection	X	X	X	√	X	X
	Abnormal Login Detection	Abnormal Login Detection	X	X	X	√	X	X
	Brute-force Attack Detection	Brute-force Attack Detection	√	√	X	√	√	X
	Lateral Movement Detection	Lateral Movement Detection	√	X	X	√	X	X
	Ransomware Detection	Ransomware Intelligence-based Detection	X	X	X	√	√ (Kernel-mode) X (User-mode)	X
		Ransomware Behavior Detection	√	√	X	√	√	X
		AI-based Ransomware Detection	√	X	X	√	X	X
	0-day or High-risk Vulnerability Detection	Vulnerability Exploitation Detection based on Abnormal Behavior	X	X	X	√	X	X
		Vulnerability Detection based on Behavior Baseline	X	X	X	√	X	X
		Stack Traceback Vulnerability Exploitation Detection	X	X	X	√	X	X

Category	Level-1 Specification	Level-2 Specification	Standard edition			Professional edition		
			Windows	Linux	macOS	Windows	Linux	macOS
		Operating System Remote RCE Vulnerability Detection	X	X	X	√	X	X
	Fileless Attack Detection	White File Exploitation Detection	X	X	X	√	X	X
		AMSI-based Script Detection	X	X	X	√	X	X
		Script Behavior Baseline Detection	X	X	X	√	X	X
		Post-remote Control Penetration Detection	X	X	X	√	X	X
		Trojan Shellcode Detection	X	X	X	√	X	X
		Process Injection Detection	X	X	X	√	X	X
	IPS Detection based on Traffic	Operating System Remote RCE Vulnerability Attack Detection	X	X	X	√	X	X
	Phishing Detection	Phishing Trojan Attachment Detection	X	X	X	√	X	X
		Phishing Webpage Detection	X	X	X	X	X	X
	Memory Scan	Real-time Memory Scan	X	X	X	√	X	X
	Remote Control Trojan Detection	Remote Control Trojan Detection	X	X	X	√	X	X
	High-performance HIPS Engine	Host Real-time Protection Engine based on Behavior Rules	√	√	√	√	√	X
	Real-time Graph Detection	Host Correlation Detection based on Real-time Graph Computing	√	√	√	√	√	X
	Data Exfiltration Detection	Ransomware Data Exfiltration Attack Detection	√	X	X	√	X	X
	Application Protection (RASP)	Supported Programming Languages	X	X	X	√	√	X
		Supported Web Frameworks	X	X	X	√	√	X
		Probe Access and Uninstallation	X	X	X	√	√	X
		Detection Rule Library Dynamic Upgrade	X	X	X	√	√	X
		Detection Capabilities	X	X	X	√	√	X
		RASP Protection Configuration	X	X	X	√	√	X
Response and Handling	Threat Event One-click Handling	File Quarantine	√	√	√	√	√	√
		Process Termination	√	√	X	√	√	X
		One-click Handling based on Process Tree	√	√	X	√	√	X
		File and Directory Deletion	√	√	√	√	√	√
		Process Execution Blocking	√	√	√	√	√	√
		Scheduled Task Cleanup	X	X	X	X	X	X
		Registry Recovery	√	NA	NA	√	NA	NA
		Office Macro Virus Recovery	√	X	X	√	X	X
		Username Quarantine	X	X	X	X	X	X
		Service Deletion	X	X	X	X	X	X
		Ransomware-encrypted File Recovery	X	X	X	√	X	X
		Virus Scan and Clean	√	√	√	√	√	√
		Network Isolation	√	√	X	√	√	X
		Response Guidance	√	√	√	√	√	√
	Threat Event Automatic Handling	Virus Event Automatic Handling	√	√	√	√	√	√
		Mining Event Automatic Handling	√	√	X	√	√	X
		Ransomware Event Automatic Handling	X	√	X	√	√	X

Category	Level-1 Specification	Level-2 Specification	Standard edition			Professional edition		
			Windows	Linux	macOS	Windows	Linux	macOS
	Multi-service Linkage Handling	Endpoint and Firewall Automatic Association	√	√	X	√	√	X
		Endpoint Linkage with Gateway to Block Attack Source	√	√	√	√	√	√
		Gateway Linkage with Endpoint to Clean Malicious Files	X	X	X	√	√	X
		Gateway Linkage with Endpoint to Terminate Malicious Outbound Processes	X	X	X	√	√	X
	Multi-product Linkage Handling	Syslog Data Outbound and Northbound Response Interface	√	√	√	√	√	√
		HiSec Insight Linkage	√	√	√	√	√	√
		LogEase Linkage	√	√	√	√	√	√
Security Analysis	Security Analysis	Event Aggregation	√	√	√	√	√	√
		Correlation Analysis	X	X	X	X	X	X
Forensics and Tracing	Alarm Automatic Tracing and Forensics	Threat Attack Path Display	X	X	X	√	√	√
	IOC Active Tracing and Forensics	IOC-targeted Active Tracing and Forensics	X	X	X	√	√	√
	Threat Hunting	Active Search for Advanced Threats	X	X	X	√	√	√
Log Reports and Notifications	Endpoint Security Notifications	Security Monthly Report	X	X	X	X	X	X
		Emergency Security Notifications	√	√	√	√	√	√
	Security Logs	Operation Log Viewing	√	√	√	√	√	√
		Operation Log Export	√	√	√	√	√	√
		Client Protection Log Export	√	√	√	√	√	√
		Threat Event Export	√	√	√	√	√	√
	Data Cleanup	Invalid Data Cleanup	√	√	√	√	√	√
Endpoint Operation and Maintenance	Operation and Maintenance	MSSP Outsourced Maintenance	√	√	√	√	√	√
	Level- and Domain-based Management	Level- and Domain-based Asset Management	√	√	√	√	√	√
	User Feedback	False Positive Feedback	√	√	X	√	√	√
	Health Status Self-check	Installation Integrity Detection	√	√	√	√	√	√
		Registry Integrity Detection	√	√	√	√	NA	√
		Service Integrity Detection	√	√	√	√	√	√
		Resident Process Number Detection	√	√	√	√	√	√
		Driver Status Detection	√	√	√	√	√	√
		Process Dump Detection	√	√	√	√	√	√
		Blue Screen Dump Detection	√	√	√	√	√	√
		Virus Scan and Clean Engine Detection	√	√	√	√	√	√
		HIPS Behavior Detection Engine Detection	√	√	√	√	√	√
		Resource Occupancy Detection	√	√	√	√	√	√
		Upgrade Failure Detection	√	√	√	√	√	√
	Fault Self-healing	Registry Entry Missing Self-healing	√	NA	NA	√	NA	√
		Configuration File Damage Self-healing	√	√	√	√	√	√
		Process File Damage Self-healing	√	√	√	√	√	√

Category	Level-1 Specification	Level-2 Specification	Standard edition			Professional edition		
			Windows	Linux	macOS	Windows	Linux	macOS
		Driver Damage	√	X	NA	√	X	NA
		Process Fault Self-healing	√	√	√	√	√	√
	Endpoint Fault Monitoring	Endpoint Fault Monitoring	√	√	√	√	√	√
	Fault Location	Agent File Forensics for Location	√	√	√	√	√	√
		Endpoint System Information Forensics for Location	√	√	√	√	√	√
	Fault Escape	Data Collection Switch	√	√	√	√	√	√
		Resource Monitoring Switch	√	√	√	√	√	√
		HIPS Engine Switch	√	√	√	√	√	√
		Graceful Degradation	√	√	√	√	√	√

(2) Network Access Control

Category	Level-1 Specification	Level-2 Specification	Standard edition			Professional edition		
			Windows	Linux	macOS	Windows	Linux	macOS
Admission Certification	Admission Certification	Certification Method	√	X	X	√	X	X
		EAP Protocol	√	X	X	√	X	X
		Certification Credentials	√	X	X	√	X	X
		CA Certificate	√	X	X	√	X	X
		Authorization Method	√	X	X	√	X	X
		Username and Password Storage	√	X	X	√	X	X
		Automatic Certification	√	X	X	√	X	X
		Release IP after Network Connection is Disconnected	√	X	X	√	X	X
		Automatically Reconnect in Case of Network Failure	√	X	X	√	X	X
		Verify Certification Server Certificate	√	X	X	√	X	X
		Proactively Log Off When the System Shuts Down or the Account is Logged Out	√	X	X	√	X	X
		Security Type and Encryption Type	√	X	X	√	X	X
		Support for SSID Black and White Lists	√	X	X	√	X	X
		Support for the Function of Forbidding Clients from Exiting	√	X	X	√	X	X
Terminal Discovery and Identification	Terminal Discovery Based on Network Traffic Analysis and Active Scanning	Terminal Identification Based on Agent Scanning	X	X	X	X	√	X
Network Security Linkage	Dynamic Baseline Inspection Linkage with Network Admission	Baseline Inspection Linkage with Network Admission	√	X	X	√	X	X
	Dynamic Threat Assessment Linkage with Network Admission	Security Threat Linkage with Network Admission	√	X	X	√	X	X
Windows Compliance Check (Posture)	Account Security Check	Password Security Check	√	X	X	√	X	X
		User Check	√	X	X	√	X	X
	Host Security Check	Operating System Version Check	√	X	X	√	X	X
		Operating System Patch Check	√	X	X	√	X	X

Category	Level-1 Specification	Level-2 Specification	Standard edition			Professional edition		
			Windows	Linux	macOS	Windows	Linux	macOS
		Patch Management Software Check	√	X	X	√	X	X
		Operating System Update Check	√	X	X	√	X	X
		Antivirus Software Check	√	X	X	√	X	X
		Firewall Check	√	X	X	√	X	X
		Disk Encryption Check	√	X	X	√	X	X
		Browser Version Check	√	X	X	√	X	X
	Device Security Check	Running Process Check	√	X	X	√	X	X
		Installed Application Check	√	X	X	√	X	X
		Running Service Check	√	X	X	√	X	X
		File Security Check	√	X	X	√	X	X
		Registry Check	√	X	X	√	X	X
		Certificate Check	√	X	X	√	X	X
		Screen Saver Check	√	X	X	√	X	X
	Non-compliance Repair	Process Repair	√	X	X	√	X	X
		Registry Repair	√	X	X	√	X	X
		Antivirus Software Repair	√	X	X	√	X	X
		Firewall Repair	√	X	X	√	X	X
		P2P Software Repair	√	X	X	√	X	X
		Patch Repair	√	X	X	√	X	X
Client Types	Permanent Client	Permanent Client	√	X	X	√	X	X
OneAgent	Unified Agent	NAC/EDR Unified Agent	√	X	X	√	X	X
Unified Management	Unified Management	Unified Configuration of NAC and Security Analysis Policies	√	X	X	√	X	X
Network-wide Linkage Response	Network-wide Linkage Response	Network+Endpoint Security Linkage Threat Response	√	X	X	√	X	X
		Network+Edge Security+Endpoint Security Linkage Threat Response Note: The purchase of perimeter security services needs to be added.	√	X	X	√	X	X

Supported Platforms

Windows PC	Windows 7 Professional, Ultimate, Enterprise (32-bit/64-bit); Windows 10 (32-bit/64-bit); Windows 11 (64-bit)
Windows Server	Windows Server 2012 R2 and above (64-bit)
Linux Server	CentOS 7 and above; EulerOS 2.0 and above; RedHat 8 and above; SUSE 12, 15; Debian 10 and above; Ubuntu 16, 18, 20, 22
macOS	macOS 11 and above